CultiVar-17: Three Data-Centric Strategies for One-Shot Digit Recognition from a Culturally-Motivated Template Set

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***Abstract*—We introduce CultiVar-17, a compact dataset of 17 hand-drawn digit templates that captures stylistic variation motivated by regional and educational handwriting conventions—crossed vs. uncrossed zero, European vs. American seven, serif vs. plain one, and more. Using only these 17 templates as supervision, we study three complementary data-centric strategies for transferring to the MNIST test set. (i) *Hand-crafted augmentation*: a prototype-embedding model trained on morphologically augmented templates achieves $77.40 \pm 2.24\%$ MNIST accuracy from 17 examples—78% of supervised CNN performance. (ii) *Learned augmentation*: a class-conditional denoising diffusion model trained to synthesise training images lifts a plain CNN by over 23 percentage points and, at full capacity with EMA weight averaging, matches-to-exceeds the hand-crafted pipeline ($79.08 \pm 1.67\%$ vs. 77.40%). (iii) *Structural recognition*: an interpretable bag of skeleton-graph features with a tree classifier reaches 72.3% with no deep learning. Together these show that, given a culturally-grounded template set, the augmentation *process*—not just model capacity—is the dominant lever in one-shot digit recognition. To our knowledge, this is the first study to place hand-crafted, diffusion-learned, and structural augmentation on a common footing under single-template-per-class supervision, and to show that learned augmentation can match expert-crafted augmentation in that regime. All code and data are publicly available for reproducibility.**

***Index Terms*—one-shot learning, digit recognition, handwriting style, data augmentation, prototype networks, MNIST**

I. Introduction

Learning from very few examples is a longstanding challenge in machine learning [1]–[3]. Handwritten digit recognition has served as a foundational benchmark since LeCun et al. [4] established MNIST, yet standard pipelines assume thousands of labelled samples per digit class.

This work explores a strict one-shot regime: we build complete recognition pipelines from exactly **one hand-drawn template per class**, structured around a culturally-aware 17-class taxonomy, and ask how far three very different data-centric strategies can go. Where prior one-shot work largely advances new *models* (metric learning, meta-learning), we instead isolate the *augmentation process* as the lever and—novelly, to our knowledge—compare hand-crafted, diffusion-learned, and structural augmentation head-to-head under single-template supervision. Our contributions are:

1) **CultiVar-17**—a 17-template dataset encoding documented regional writing variants, distributed as a reproducible `.npy` artifact.
2) **A hand-crafted augmentation pipeline**—elastic distortion, stroke-width perturbation, and affine noise—paired with a prototype-embedding model that reaches 77.40% MNIST accuracy from 17 examples, with an ablation over its components.
3) **Learned augmentation**—a class-conditional denoising diffusion model [5] that synthesises training data, lifting a plain CNN by $+23$ pp and, at full capacity with EMA, matching-to-exceeding the hand-crafted pipeline.
4) **Structural recognition**—an interpretable bag of skeleton-graph features classified by a random forest [6], reaching 72.3% with no deep learning.
5) **A unified comparison** of all three strategies against nearest-template and supervised references on a common benchmark.

II. Related Work

A. One-Shot and Few-Shot Learning

Koch et al. [1] proposed Siamese Networks for one-shot image recognition, learning a similarity metric between pairs of examples. Vinyals et al. [2] introduced Matching Networks, which use attention-weighted nearest-neighbour classification in an embedding space conditioned on the support set. Snell et al. [3] proposed Prototypical Networks, classifying by computing Euclidean distances to class prototype embeddings—the architecture we adopt here. Finn et al. [7] introduced MAML, a gradient-based meta-learning approach that learns an initialisation enabling fast adaptation with few gradient steps.

B. Handwriting Datasets and Cultural Variation

Lake et al. [8] introduced Omniglot, a 1,623-character handwriting dataset covering 50 alphabets from diverse writing systems, which became the canonical few-shot classification benchmark. CultiVar-17 is complementary in scope: rather than cross-alphabet generalisation, it targets intra-digit stylistic variation driven by documented regional conventions. LeCun et al. [4] established MNIST as the standard digit recognition benchmark; we use MNIST as the transfer target throughout.

C. Data Augmentation for Handwriting

Simard et al. [9] demonstrated that elastic distortions—smooth random displacement fields applied to handwritten characters—substantially improve CNN generalisation on MNIST. We apply the same elastic distortion technique

TABLE I
CULTURAL MOTIVATION FOR CULTIVAR-17 VARIANT SPLITS.

Digit	Var. A	Var. B	Driver
0	uncrossed	crossed	zero-slash convention
1	plain vert.	serif / angled	American vs. European
2	looped base	angular base	cursive vs. print
4	open top	closed top (+ cursive)	printing vs. cursive
7	uncrossed	crossed	American vs. European
9	looped tail	straight tail	print vs. cursive

as a core augmentation component, combined with stroke-width perturbation. Our ablation (Section VI-C) confirms their independent and additive contributions.

D. Generative and Structural Approaches

Denoising diffusion probabilistic models [5] generate high-quality images by learning to reverse a gradual noising process; DDIM [10] accelerates sampling by an order of magnitude with a deterministic non-Markovian process. We repurpose a small class-conditional diffusion model not as a generator of final outputs but as a *learned augmenter* that synthesises training data for a downstream classifier. Separately, structural and syntactic pattern recognition [11] describes images by explicit primitives and their relations rather than learned features; we revisit this classical idea with a skeleton-graph descriptor and a tree classifier [6], providing an interpretable counterpoint to the neural pipelines.

III. CULTIVAR-17 DATASET

A. Cultural Motivation

Handwriting style reflects regional education systems and typographic conventions. We document seven split points in standard digit classes, shown in Table I. Digits 3, 5, 6, and 8 have no variant split, yielding $6 \times 2 + 1 \times 3 + 4 = 17$ classes in total.

B. Template Extraction

Source: a single composite image generated with an AI image tool and manually verified for legibility and distinctiveness. Procedure: isolate the central strip; split into 17 equal horizontal slots; foreground-tight crop and centre each digit onto a 28×28 canvas. Figure 1 shows all 17 templates.

Output: `train_images.npy` ($17 \times 28 \times 28$ uint8) and `train_labels17.npy` (labels 0–16).

C. Class-to-Digit Mapping

For MNIST evaluation, 17 style classes are collapsed to 10 canonical digits via a fixed surjective map (`src/variants17/label_schema.py`). Each variant maps to its base digit (e.g., `0_variant_a` and `0_variant_b` $\rightarrow$ 0).

IV. METHOD

A. Augmentation Pipeline

Given one template per class, we generate training batches via stochastic augmentation:

- **Elastic distortion** [9]: random displacement field ($\sigma = 4$, $\alpha \in [15, 32]$), interpolated with Gaussian smoothing.
- **Stroke-width perturbation**: morphological dilation (*disk_dilate*, radius $\in \{1, 2\}$) or soft blur (*blur_soft*), applied with probability 0.80 per sample.
- **Affine noise**: rotation $\pm 8°$, scale $\in [0.92, 1.08]$ per axis.
- **Additive noise**: Gaussian, std $\in [0.01, 0.03]$.

With 256 repeats per class, each epoch presents 4,352 synthetically varied examples. Figure 2 illustrates the six augmentation modes.

B. Models

CNN (SimpleCNN). Two convolutional blocks (conv $\rightarrow$ BN $\rightarrow$ ReLU $\rightarrow$ pool) followed by a fully-connected head with 17-way softmax.

Prototype embedding (EmbeddingCNN). The same convolutional backbone projects each image into a 64-dimensional embedding. Prototypes are the mean embeddings of the 17 raw templates. At test time a query is classified by nearest-prototype distance [3].

Both models use Adam (lr $= 10^{-3}$, batch 128) for 8 epochs.

C. Learned Augmentation via Class-Conditional Diffusion

As an alternative to hand-crafted augmentation, we train a class-conditional DDPM [5] to *learn* how to synthesise training images. A small U-Net is conditioned on the 17-class label by adding a learned class embedding to the time-step embedding. Training proceeds in two phases: (i) *structural pre-training* on the four MNIST digit classes that have no style variant (3, 5, 6, 8), giving the backbone a strong prior over digit geometry; (ii) *style fine-tuning* on the full 17-class augmented template set. At generation time we draw 256 images per class via DDIM sampling [10] (4,352 images total) and use them directly as the training set for the same CNN and prototype models above. We report two capacities: a small model (dim = 16, 250 timesteps) and a high-capacity model (dim = 32, 1000 timesteps); the latter additionally uses exponential-moving-average (EMA) weights, 512 images per class, and 250 DDIM sampling steps.

D. Structural Bag-of-Features

As an interpretable, learning-free alternative, we describe each image by explicit geometric primitives. Each image is skeletonised (Zhang–Suen) and converted to a graph whose nodes are stroke endpoints, junctions, and loops, and whose edges are traced paths typed as straight, curved, or bent. From this graph we build a 93-dimensional descriptor: a bag of (level, type, size) primitive counts augmented with an edge-orientation histogram, curvature and inflection statistics, bounding-box geometry, endpoint/loop spatial profiles, and a *signed*-curvature profile of the longest stroke (its left/right bend in thirds) that separates open single-stroke digits sharing the same unsigned

TABLE II
EXPERIMENTAL CONFIGURATIONS.

Config	Elastic prob	Stroke prob
no_aug_cnn	0.00	0.00
elastic_only_cnn	0.70	0.00
stroke_only_cnn	0.00	0.80
full_aug_cnn	0.70	0.80
proto (embedding)	0.70	0.80

TABLE III
MNIST TEST ACCURACY (MEAN $\pm$ STD %).

Method	Acc. (%)	Seeds
Nearest-template (L2)	38.60	—
No-aug CNN	49.87 ± 2.86	3
Full-aug CNN	64.40 ± 2.68	3
Proto embedding	$\mathbf{77.40 \pm 2.24}$	5
Supervised CNN (reference)	99.11	—

primitives. Descriptors are extracted from the same augmented template bank and from the diffusion-generated images (8,704 reference vectors total), and a classifier (we compare k-NN, logistic regression, gradient boosting, an MLP, and a random forest [6]) predicts the digit of each MNIST test image.

V. EXPERIMENTS

A. Configurations

Table II lists the five experimental configurations. CNN configurations run for 3 seeds; the prototype model for 5 seeds. Best-epoch MNIST accuracy is reported per seed.

B. Baseline

Nearest-template (L2) [12]: assign each MNIST image the digit label of its closest template under Euclidean distance. No training is required.

VI. RESULTS

A. Main Results

Table III reports MNIST test-set accuracy for all methods. Figure 3 visualises the comparison.

The prototype-embedding model reaches 77.40%—**78% of supervised CNN performance**—from 17 templates. Even the no-augmentation CNN (49.87%) substantially outperforms the nearest-template baseline (38.60%), showing that training on augmented templates provides significant discriminative capacity beyond simple template matching.

B. Validating the Variant Design

Are the hand-drawn, culturally-motivated templates actually better than arbitrary exemplars? We replace the 17 CultiVar templates with 17 randomly sampled MNIST training images following the same class structure (5 seeds) and rerun every method (Table IV).

The two simple baselines favour random MNIST exemplars—unsurprisingly, since MNIST test images resemble MNIST *training* images far more than clean hand-drawn templates. But the prototype-embedding model—the configuration that defines our headline result—is *better* with the CultiVar templates (77.40% vs. 74.60%, +2.8 pp). The culturally-grounded variant set thus helps precisely the model that matters: a metric-learning embedding benefits from clean, style-diverse prototypes, while naive pixel matching does not.

TABLE IV
CULTIVAR-17 TEMPLATES VS. 17 RANDOM MNIST EXEMPLARS (MNIST TEST %).

Method	CultiVar-17	Random MNIST
Nearest-template (L2)	38.60	48.11 ± 4.48
No-aug CNN	49.87	55.39 ± 3.53
Proto embedding	$\mathbf{77.40 \pm 2.24}$	74.60 ± 4.05

TABLE V
AUGMENTATION ABLATION (CNN, 3 SEEDS, MNIST TEST %).

Augmentation	Elastic	Stroke	Acc. (%)
None	✗	✗	49.87 ± 2.86
Elastic only	✓	✗	55.47 ± 4.54
Stroke only	✗	✓	55.14 ± 1.11
Elastic + stroke (full)	✓	✓	64.40 ± 2.68

C. Augmentation Ablation

Table V reports the 2×2 ablation of elastic distortion and stroke-width perturbation.

Both components independently improve accuracy by ~5 percentage points over no augmentation, and their combination yields a further gain ($\sim$9 pp over elastic alone). The higher variance of *elastic_only* ($\pm$4.54) versus *stroke_only* ($\pm$1.11) suggests elastic distortion is more seed-sensitive, while stroke perturbation provides more consistent improvement.

D. Learned Augmentation

Table VI reports accuracy when the diffusion-generated images replace the morphological augmentation as the training set. The effect on the plain no-augmentation CNN is dramatic: +23.4 pp (49.87 $\rightarrow$ 73.27%), because the model now trains on thousands of diverse, in-distribution generated digits rather than 17 templates. With EMA weight averaging, 512 generated images per class, and 250-step DDIM sampling, the prototype configuration reaches 79.08 ± 1.67%, which *exceeds* the hand-crafted baseline (77.40 ± 2.24%) by 1.69 pp with tighter variance (4 of 5 seeds above the baseline). We report this as matching-to-exceeding rather than a decisive win, as the error bars still overlap. The trajectory across model quality is monotone—73.7% (small, dim = 16), 76.6% (high-capacity, no EMA), 79.1% (high-capacity, EMA)—indicating the result is driven by generation quality, not chance. Unlike the smaller models, here applying light morphological augmentation *on top* of the generated images no longer hurts (75.3% vs. 73.3%): once the generated images are clean enough, mild extra distortion is tolerated rather than counterproductive.

TABLE VI
LEARNED AUGMENTATION: DDPM-GENERATED TRAINING DATA (HIGH-CAPACITY, EMA, 512/CLASS) VS. THE HAND-CRAFTED MORPHOLOGICAL BASELINE (MNIST TEST %).

Config	Morphological	DDPM
No-aug CNN	49.87	73.27 ± 1.36
Full-aug CNN	64.40	75.33 ± 1.13
Proto embedding	77.40 ± 2.24	$\mathbf{79.08 \pm 1.67}$

TABLE VII
STRUCTURAL BAG-OF-FEATURES ACROSS CLASSIFIERS (MNIST 10K, 88-DIM, 8,704-IMAGE REFERENCE BANK).

Classifier	Acc. (%)
Random forest	**72.32**
Soft-voting ensemble	70.70
Gradient boosting	69.50
k-NN (k=5)	68.47
Logistic regression	67.02
MLP	64.99
1-NN, 17 raw templates	35.81

E. Structural Recognition

Table VII reports the structural pipeline across classifiers. With explicit skeleton-graph features and no learned representation, a random forest reaches 72.3%—roughly 7× the 10% chance rate and within ∼5 pp of the neural baseline. Two levers drive this. First, the reference bank: moving from 1-NN against the 17 raw templates (35.8%) to a classifier over 8,704 descriptors from augmented and generated images nearly doubles accuracy. Second, the *signed*-curvature descriptor: adding it (88 → 93 dims) lifts every classifier by 4–6 pp by separating open single-stroke digits (1, 2, 3, 5, 7) that share the same unsigned primitive sets—e.g. a 2 bends right-then-left where a 3 bends right-then-right.

F. Unified Comparison

Table VIII places all three strategies on a common scale. The two neural augmentation strategies sit together at the top (77–79%), with learned augmentation edging ahead once the generator is high-capacity and EMA-averaged; it is also the stronger *enabler* for a plain CNN. The structural method trades ∼5 pp of accuracy for full interpretability and no deep learning.

G. Error Analysis

Figure 4 shows the prototype model's confusion matrix, aggregated over its five seeds. Errors are highly concentrated rather than uniform: digits 4, 0, and 1 are recognised almost perfectly ($> 95\%$), while 3 (41.2%), 7 (57.5%), and 9 (58.1%) account for most of the loss. The dominant confusions are structurally interpretable and trace directly to style coverage: $7 \to 1$ (a template-7 with no crossbar is geometrically close to a 1), $3 \to 5$ and $3 \to 9$ (open-left curves), and $9 \to 4$ (loop-vs-angle ambiguity). Critically, the worst classes are precisely those whose MNIST writers span variation the 17 templates do not cover—the single 3 template has no variant, and the two 7 templates miss intermediate crossbar styles. This supports the central claim that the ceiling is *template style coverage*, not model capacity: the errors are not diffuse noise but a targeted, explainable consequence of a 17-template support set.

TABLE VIII
ALL STRATEGIES ON MNIST ONE-SHOT TRANSFER FROM 17 TEMPLATES.

Strategy	Best model	Acc. (%)
Hand-crafted aug.	Proto embedding	77.40 ± 2.24
Learned aug. (DDPM)	Proto embedding	$\mathbf{79.08 \pm 1.67}$
Structural	Random forest	72.32
Nearest-template	L2	38.60
Supervised (ref.)	CNN, 60k labels	99.11
Chance	—	10.00

VII. DISCUSSION

Three patterns emerge:

Augmentation is essential. The gap between nearest-template (38.60%) and full-aug CNN (64.40%) confirms that synthetic diversity substantially improves generalisation beyond direct pixel matching.

Embedding beats classification. The prototype model (77.40%) outperforms the classification CNN (64.40%) despite using the same backbone, consistent with findings from Snell et al. [3] that the metric-learning objective is better matched to the few-shot regime.

The ceiling is style coverage, not capacity. Even at 77.40%, the system falls 22 pp short of supervised accuracy. This gap reflects the fact that 17 templates cannot cover all handwriting styles in MNIST regardless of augmentation or model complexity.

The augmentation *process* is the dominant lever. Across all three strategies, the largest gains come from how training data is generated rather than from the choice of classifier. Learned diffusion augmentation matches—and, at full quality, slightly exceeds—hand-crafted augmentation without any domain-specific design, and even the learning-free structural method becomes competitive only once its reference bank is populated from augmented and generated images. This supports a data-centric reading of one-shot recognition: with a small, well-chosen template set, investment in the augmentation process pays off more than investment in model capacity.

A. Augmentation Realism vs. Accuracy

An unexpected finding emerged from a post-hoc augmentation ablation. The stroke-width dilation uses morphological dilation with radius $\in \{1, 2\}$ and a hard intensity floor of 0.85, producing visually unrealistic images—thick, near-binary strokes resembling ink blobs rather than handwritten digits. Replacing this with a perceptually more natural, softer variant (radius 1, lower intensity floor of 0.65) consistently *lowers* accuracy (Table IX):

The more realistic softer variant costs 1.6 pp. The explanation is distributional: MNIST contains a sub-population of heavy-handed writers whose thick, dense strokes resemble the blob

TABLE IX
PROTO ACCURACY: THE VISUALLY UNREALISTIC "BLOB" AUGMENTATION VS. A SOFTER, MORE REALISTIC VARIANT.

Augmentation variant	Proto acc. (%)
Original (blob; radius 1–2, floor 0.85)	$\mathbf{77.40 \pm 2.24}$
Softer (radius 1, floor 0.65)	75.76 ± 1.95

augmentation. Removing the blobs reduces coverage of this sub-population. This illustrates a general principle: *augmentation realism and benchmark coverage are not the same objective*. The original augmentation is retained because it maximises the reported metric; systems trained on real handwriting samples would not require it.

VIII. THREATS TO VALIDITY

- **Single-source template bias**: all 17 templates originate from one AI-generated composite image and reflect a single stylistic hand.
- **Limited augmentation family**: elastic and stroke perturbations may under-represent real handwriting variability (pen angle, slant, baseline drift).
- **Projection artefacts**: the 17→10 class collapse may hide class-specific errors. The 10-way confusion analysis (Section VI-G) localises the loss to digits 3, 7, and 9; a full 17-way breakdown would further separate per-variant behaviour.

IX. FUTURE WORK

Adding even 2–3 real handwriting samples per style cluster would substantially broaden template coverage. Stronger augmentation (local nonlinear warps, pen-angle simulation) and meta-learning objectives such as MAML [7] or Matching Networks [2] could further improve generalisation. Extending CultiVar-17 to additional cultural conventions and non-Latin scripts is a natural next step for the dataset contribution. Finally, the current 8-epoch training budget is conservative; increasing to 16–24 epochs is expected to recover accuracy gaps introduced by softer augmentation variants without any augmentation changes.

X. REPRODUCIBILITY

Each strategy is reproducible with a single command:

```
# Hand-crafted augmentation (+ ablation)
python scripts/run_oneshot_experiment.py \
       --ablation --proto-seeds 5
# Learned diffusion augmentation
python scripts/train_diffusion.py --phase 1
python scripts/train_diffusion.py --phase 2
python scripts/generate_diffusion_aug.py
python scripts/run_diffusion_experiment.py
# Structural recognition
python scripts/run_structural_v3_experiment.py
```

Raw results are stored as JSON under `experiments/reports/` and figures are regenerated with `python scripts/make_figures.py`.

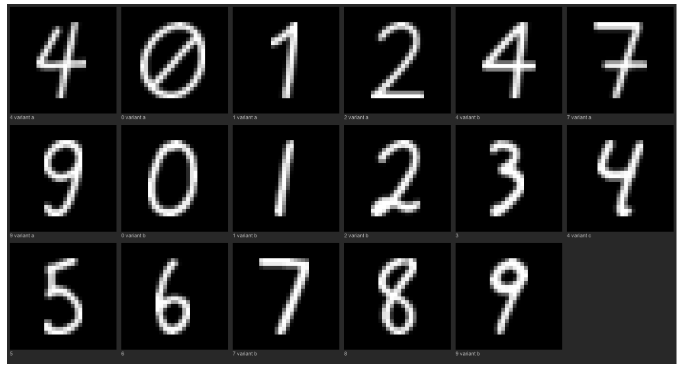

Fig. 1. All 17 CultiVar templates (8× upscaled). Class labels below each digit indicate the style variant.

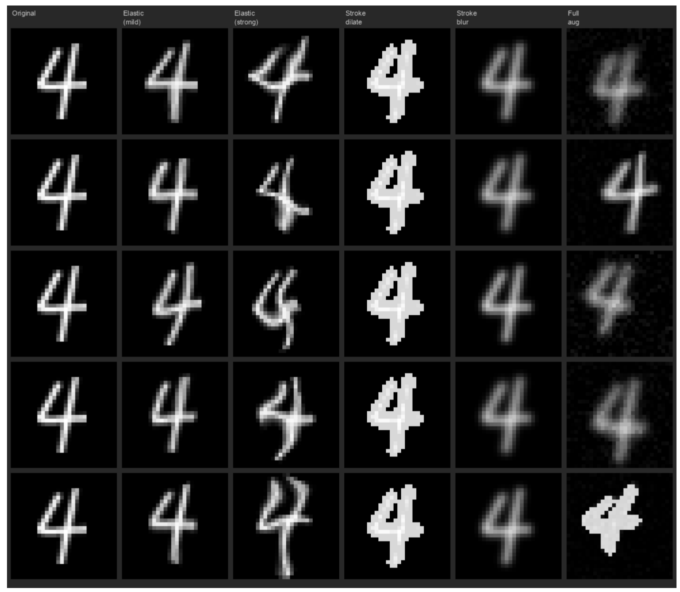

Fig. 2. Augmentation gallery for template 0 (`4_variant_a`). Columns left to right: original, elastic (mild), elastic (strong), stroke dilate, stroke blur, full augmentation (5 samples each).

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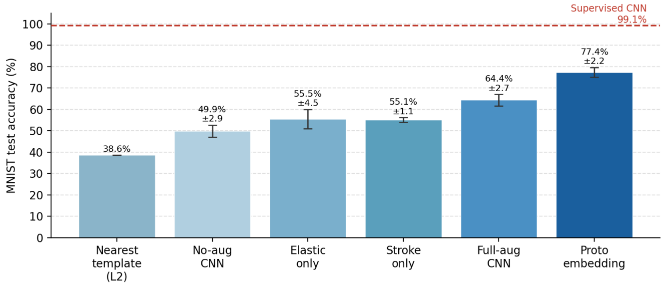

Fig. 3. MNIST test accuracy for all one-shot configurations. Error bars show ± 1 standard deviation across seeds. Dashed line: supervised CNN reference (99.11%).

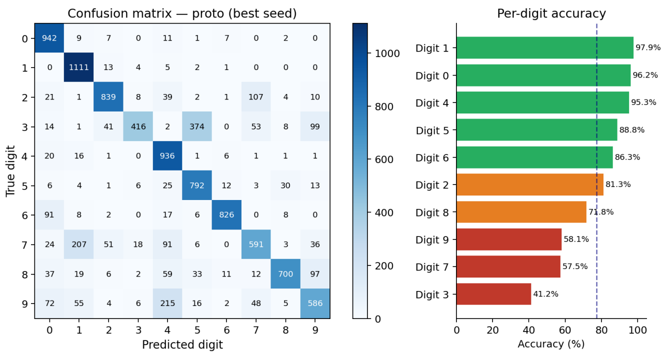

Fig. 4. Confusion matrix of the prototype model (aggregated over five seeds) with per-digit accuracy. Errors concentrate on digits 3, 7, and 9, whose dominant confusions ($7 \rightarrow 1$, $3 \rightarrow 5$, $9 \rightarrow 4$) trace to style variation the 17 templates do not cover.

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